

of other pieces of hardware—usually a system in the design stage. Developers can use an emulator in place of a processor or chip that is being built, and run the system with live data to see how it works. They can watch the application run step-by-step and easily identify errors. Emulators are a very important tool in functional verification and debugging of embedded systems where simulation can be slow and costly.

One of the most popular types of hardware emulators used in embedded design and development is the in-circuit emulator, which is used to test embedded software. Basically, it allows the developers to test the embedded software on the target hardware in order to ensure that there is no faulty code.

Earlier, emulators were independent electronic devices that could be plugged into a board in place of the microprocessor or other chips, to emulate their behaviour for testing purposes. However, the current generation of emulators allows the users to test their code on the actual target system itself using special Joint Test Action Group (JTAG) based interfaces.

Companies such as Cadence Design Systems and Mentor Graphics even have emulation machines amongst their portfolio. These machines have a board with a very large number of FPGAs, and the company provides software using which a user can simulate the design, with the aid of the hardware. How is this done? Simulation of a design typically takes machine cycles, when you run it on

your Linux or PC box. However, for complex modules, it is probably way faster if you synthesise it, load it on an FPGA and run simulation vectors through it. This way, they simulate with the help of hardware, and achieve much faster verification times. One big disadvantage of this method is the high cost; these solutions might cost millions of dollars.

FPGAs come of age

An FPGA is essentially a chip that can be configured for various purposes by the user. As a result, it can be used to prototype system-on-a-chip, ASIC and embedded system designs. These prototypes, in turn, are used for functional verification of the hardware design, as well as the software and firmware being developed for it. In many cases, FPGA-based functional verification is considered to be more accurate than software-based simulation.

Earlier, it was believed that FPGA-based prototyping and verification is very cumbersome and time-consuming for large systems. However, the situation has changed now. New-age FPGAs are used extensively to test and prototype large modules of designs. These can even handle multi-million gate designs. A single high-performance ASIC prototyping system (HAPS) board from Synopsis, for example, can hold up to 8 million ASIC gates. More recent solutions from Synopsis can actually break a large full chip design into partitions, load these partitions onto a multi-FPGA board and prototype a full system-on-a-chip in one go. This was not possible until a few years

ago when engineers had to manually partition and prototype the design bottom up.

Apart from identifying faults and testing or debugging software developed for embedded systems, FPGA-based prototypes have also been found to be useful in kindling customer interest by showing them a close-to-real demonstration of the system's features.

Xilinx and Altera are the market leaders in the FPGA space, while companies such as Synopsis and Mentor Graphics provide software-based tools for the design and implementation of FPGAs.

The two tools discussed above, namely, emulators and FPGA-based prototypes, also help in another key area of verification—hardware-software integration. Suppose you have an early version of your design (say, a register transfer level description). You prototype it and load to your FPGA. Using some system-level design tools and languages such as System C (which have the capability of modelling systems using transactions; also called transaction-level models), you can test even the device driver software. Engineers can run predetermined tests using embedded real-time operating systems (RTOS). More than checking the performance of the system (remember, there is just a prototype, not the final chip design), one can test functional correctness of the design very accurately.

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